

Exercise 1: SQL Fundamentals SELECT & Filtering

01. Retrieve all columns from the employees table

Run selected (1000) ✓ 1 minute ago (1s) workspace: default

```
16 );
17
18 --Step 4: Load the information
19 INSERT INTO company.sales.employees VALUES
20 (1,'Alice','Green','IT',70000,'2020-01-10','Johannesburg'),
21 (2,'Brian','Lee','HR',45000,'2019-03-22','Cape Town'),
22 (3,'Cathy','Zulu','Finance',65000,'2018-07-18','Durban'),
23 (4,'David','Mokoena','Marketing',50000,'2021-11-05','Pretoria'),
24 (5,'Eva','Naidoo','IT',72000,'2017-09-30','Johannesburg');
25
26 --Step 5: Query the table
27 SELECT* FROM company.sales.employees;
28
29 --Question 1: Retrieve all columns from the employees table.
30 SELECT*
31 FROM company.sales.employees;
32
```

Add parameter

Table +

	id	first_name	last_name	department	salary	hire_date	city
1	4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
2	1	Alice	Green	IT	70000	2020-01-10	Johannesbu...
3	2	Brian	Lee	HR	45000	2019-03-22	Cape Town
4	3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
5	5	Eva	Naidoo	IT	72000	2017-09-30	Johannesbu...

5 rows | 0.998s runtime | See performance (1) | Checked for improvements

Refreshed 1m ago

02. Find all unique departments:

Run selected (1000) ✓ Just now (1s) workspace: default

```
22 (3,'Cathy','Zulu','Finance',65000,'2018-07-18','Durban'),
23 (4,'David','Mokoena','Marketing',50000,'2021-11-05','Pretoria'),
24 (5,'Eva','Naidoo','IT',72000,'2017-09-30','Johannesburg');
25
26 --Step 5: Query the table
27 SELECT* FROM company.sales.employees;
28
29 --Question 1: Retrieve all columns from the employees table.
30 SELECT*
31 FROM company.sales.employees;
32
33 --Question 2: Find all unique departments.
34
35 SELECT DISTINCT department
36 FROM company.sales.employees;
37
```

Add parameter

Table +

	department
1	Marketing
2	IT
3	HR
4	Finance

4 rows | 0.585s runtime | See performance (1) | Checked for improvements

Refreshed Now

Boitumelo Khumalo

Data Analytics

BrightLearn

03. Retrieve first and last names ordered by salary descending

Run selected (1000) ✓ Just now (1s) workspace default

Edit

```
27 SELECT* FROM company.sales.employees;
28
29 --Question 1: Retrieve all columns from the employees table.
30 SELECT*
31 FROM company.sales.employees;
32
33 --Question 2:Find all unique departments.
34
35 SELECT DISTINCT department
36 FROM company.sales.employees;
37
38 --Question 3: Retrieve first and last names ordered by salary descending
39
40 SELECT first_name,last_name,salary
41 FROM company.sales.employees
42 ORDER BY salary DESC;
43
```

Add parameter

Table +

	A _C first_name	A _C last_name	I ₃ salary
1	Eva	Naidoo	72000
2	Alice	Green	70000
3	Cathy	Zulu	65000
4	David	Mokoena	50000
5	Brian	Lee	45000

↓

 5 rows | 0.530s runtime [See performance \(1\)](#) ✓ Checked for improvements Refreshed Now

04. Retrieve the top 3 highest paid employees

Run selected (1000) ✓ 2 minutes ago (1s) workspace default

Edit

```
34
35 SELECT DISTINCT department
36 FROM company.sales.employees;
37
38 --Question 3: Retrieve first and last names ordered by salary descending
39
40 SELECT first_name,last_name,salary
41 FROM company.sales.employees
42 ORDER BY salary DESC;
43
44 --Question 4: Retrieve the top 3 highest paid employees
45
46 SELECT first_name,last_name,salary
47 FROM company.sales.employees
48 ORDER BY salary DESC
49 LIMIT 3;
50
```

Add parameter

Table +

	A _C first_name	A _C last_name	I ₃ salary
1	Eva	Naidoo	72000
2	Alice	Green	70000
3	Cathy	Zulu	65000

↓

 3 rows | 0.533s runtime [See performance \(1\)](#) ✓ Checked for improvements Refreshed 2m ago

Boitumelo Khumalo

Data Analytics

BrightLearn

05. Find employees in the IT department:

▶ Run selected (1000) ✓ Just now (<1s) workspace default

⋮ Edit

●

📅

📄

💬

📤

```
40 SELECT first_name,last_name,salary
41 FROM company.sales.employees
42 ORDER BY salary DESC;
43
44 --Question 4: Retrieve the top 3 highest paid employees
45
46 SELECT first_name,last_name,salary
47 FROM company.sales.employees
48 ORDER BY salary DESC
49 LIMIT 3;
50
51 --Question 5: Find employees in the IT department
52
53 SELECT first_name,last_name,department
54 FROM company.sales.employees
55 WHERE department = 'IT';
56
```

Add parameter

Table +

	A ^B _C first_name	A ^B _C last_name	A ^B _C department
1	Alice	Green	IT
2	Eva	Naidoo	IT

⬇

2 rows

0.348s runtime

[See performance \(1\)](#)

✓ Checked for improvements

Refreshed Now

06. Find employees in finance with salary > 60000

▶ Run selected (1000) ✓ Just now (<1s) workspace default

⋮ Edit

●

📅

📄

💬

📤

```
45
46 SELECT first_name,last_name,salary
47 FROM company.sales.employees
48 ORDER BY salary DESC
49 LIMIT 3;
50
51 --Question 5: Find employees in the IT department
52
53 SELECT first_name,last_name,department
54 FROM company.sales.employees
55 WHERE department = 'IT';
56
57 --Question 6: Find employees in finance with salary > 60000
58
59 SELECT first_name,last_name,salary,department
60 FROM company.sales.employees
61 WHERE department = 'Finance' AND salary > 60000;
```

Add parameter

Table +

	A ^B _C first_name	A ^B _C last_name	1 ² ₃ salary	A ^B _C department
1	Cathy	Zulu	65000	Finance

⬇

1 row

0.323s runtime

[See performance \(1\)](#)

✓ Checked for improvements

Refreshed Now

Boitumelo Khumalo

Data Analytics

BrightLearn

07. Find employees in HR or Marketing:

▶ Run selected (1000) ✓ Just now (1s) workspace default

Edit

```
51 --Question 5: Find employees in the IT department
52
53 SELECT first_name,last_name,department
54 FROM company.sales.employees
55 WHERE department = 'IT';
56
57 --Question 6:Find employees in finance with salary > 60000
58
59 SELECT first_name,last_name,salary,department
60 FROM company.sales.employees
61 WHERE department = 'Finance' AND salary > 60000;
62
63 --Question 7: Find employees in HR or Marketing:
64
65 SELECT first_name,last_name,department
66 FROM company.sales.employees
67 WHERE department = 'HR' OR department = 'Marketing';
```

Add parameter

Table +

	first_name	last_name	department
1	David	Mokoena	Marketing
2	Brian	Lee	HR

2 rows | 0.664s runtime See performance (1) ✓ Checked for improvements

Refreshed Now

08. Find employees not in IT:

▶ Run selected (1000) ✓ Just now (11s) workspace default

Edit

```
56
57 --Question 6:Find employees in finance with salary > 60000
58
59 SELECT first_name,last_name,salary,department
60 FROM company.sales.employees
61 WHERE department = 'Finance' AND salary > 60000;
62
63 --Question 7: Find employees in HR or Marketing:
64
65 SELECT first_name,last_name,department
66 FROM company.sales.employees
67 WHERE department = 'HR' OR department = 'Marketing';
68
69 --Question 8: Find employees not in IT:
70 SELECT first_name,last_name,department
71 FROM company.sales.employees
72 WHERE department != 'IT';
```

Add parameter

Table +

	first_name	last_name	department
1	David	Mokoena	Marketing
2	Brian	Lee	HR
3	Cathy	Zulu	Finance

3 rows | 10.981s runtime See performance (1) ✓ Checked for improvements

Refreshed Now

Boitumelo Khumalo

Data Analytics

BrightLearn

09. Find employees in IT, HR, or Finance using IN:

▶ Run selected (1000) ✓ Just now (1s) workspace default

⋮ Edit

```
62
63 --Question 7: Find employees in HR or Marketing:
64
65 SELECT first_name,last_name,department
66 FROM company.sales.employees
67 WHERE department = 'HR' OR department = 'Marketing';
68
69 --Question 8: Find employees not in IT:
70 SELECT first_name,last_name,department
71 FROM company.sales.employees
72 WHERE department != 'IT';
73
74 --Question 9: Find employees in IT, HR, or Finance using IN:
75
76 SELECT first_name,last_name,department
77 FROM company.sales.employees
78 WHERE department IN ('IT','HR','Finance');
```

Add parameter

Table +

	A ^B _C first_name	A ^B _C last_name	A ^B _C department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	Eva	Naidoo	IT

↓ 4 rows 0.886s runtime [See performance \(1\)](#) ✓ Checked for improvements Refreshed Now

10. Find employees in IT with salary > 65000 and city Johannesburg:

▶ Run selected (1000) ✓ Just now (<1s) workspace default

⋮ Edit

```
68
69 --Question 8: Find employees not in IT:
70 SELECT first_name,last_name,department
71 FROM company.sales.employees
72 WHERE department != 'IT';
73
74 --Question 9: Find employees in IT, HR, or Finance using IN:
75
76 SELECT first_name,last_name,department
77 FROM company.sales.employees
78 WHERE department IN ('IT','HR','Finance');
79
80 --Question 10: Find employees in IT with salary > 65000 and city Johannesburg:
81
82 SELECT first_name,last_name,salary,department,city
83 FROM company.sales.employees
84 WHERE department = 'IT' AND salary > 65000 AND city = 'Johannesburg';
```

Add parameter

Table +

	A ^B _C first_name	A ^B _C last_name	1 ² ₃ salary	A ^B _C department	A ^B _C city
1	Alice	Green	70000	IT	Johannesbu...
2	Eva	Naidoo	72000	IT	Johannesbu...

↓ 2 rows 0.403s runtime [See performance \(1\)](#) ✓ Checked for improvements Refreshed Now